

How to generate images with stable diffusion?

We will now use a script called `txt2img.py` to convert text prompts into 512x512 images. To check if everything is working correctly, you can try the following example command in your console:

```
• python scripts/txt2img.py --prompt "a close-up portrait of a cat by pablo picasso, vivid, abstract art, colorful, vibrant" --plms --n_iter 5 --n_samples 1
```

Your console will show a progress indicator as it produces the images. This command will generate five images of cats, which will be saved at `C:\stable-diffusion\stable-diffusion-main\outputs\txt2img-samples\samples`.

To generate a different image, all you need to do is change the text in the double-quotation marks following `--prompt`. For example, to generate a realistic image of a gopher wearing a wizard's hat in a magical forest, you can use the following command:

```
• python scripts/txt2img.py --prompt "a photograph of a gopher wearing a wizard hat in a forest, vivid, photorealistic, magical, fantasy, 8K UHD, photography" --plms --n_iter 5 --n_samples 1
```

Make sure to include specific terms related to your desired image. If you want a photorealistic image, specify that in the prompt. If you want an image inspired by a specific artist, include the artist's name in the prompt.

What do the arguments in the command mean?

The command used to generate images in Stable Diffusion comes with several arguments that can be customized to obtain the desired results. Here are some of the most important ones required to run Stable Diffusion on an average gaming computer:

- **--plms:** This argument specifies how the images will be sampled. If you're interested in the math behind this, you can read the related paper.
- **--n_iter:** This argument determines the number of iterations for each prompt. A good starting point is 5 iterations to see the kind of results you're getting.
- **--n_samples:** This argument sets the number of images that will be generated. By default, it's set to 3, but this might cause issues with the VRAM of most computers. It's recommended to stick to 1 unless you have a specific reason to increase it.

Aside from these, there are several other arguments that you can use to